

SEUNGMIN HAN

Systems Architect · Capital Markets & Digital Assets · AI-Augmented Engineering

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PROFILE

An advanced engineer who became an architect. I spent years building the low-level systems securities firms trade on; now I design them — and I know what separates a **good system** from a fragile one: reversibility, legibility, fail-closed safety, and minimal standing state. Increasingly I pair that judgment with AI, directing agentic tooling to architect and ship large systems well beyond one engineer's throughput. My foundation runs deep — market-making and execution infrastructure at Korea Investment Securities, plus applied-cryptography research (fully homomorphic encryption; two patents) directly relevant to **digital-asset privacy, custody, and settlement**. I work across engineering, product, and business in Korean and English, focused on the build-out of global tokenized- and digital-asset market infrastructure.

PROFESSIONAL EXPERIENCE

Korea Investment Securities (한국투자증권) — Investment Engineering Dept. · Seoul Sep 2024 – Present

Market-Making / Liquidity-Provision (MM/LP) Systems Developer

- Sole developer-owner of the firm's MM/LP trading platform, working **full-stack across the entire trading stack** — trader-facing UI (C# / DevExpress WinForms), back-end services (legacy C# and new C++), the data & analytics layer, and the underlying hardware / infrastructure — quoting across **~5,500 instruments simultaneously** over a Korea Exchange (KRX) connectivity stack.
- VEGA** — architected and am delivering a **strangler-fig re-architecture** of the firm's 13-service .NET trading monolith into independently deployable services, migrating risk-critical order logic behind a legacy "parity oracle" that proves behavior is preserved (never a big-bang cutover).
- NOVA** — built from scratch a **C++20 ultra-low-latency order/hedge engine** (tick-to-trade, single-digit-μs target): core-pinned threads on a zero-hop hot path, zero-copy parsing, lock-free quote tables, and a shared-memory parameter plane — the basis of a solo effort to migrate the entire **Windows + C# platform to Linux + C++**.
- Designed, shipped, and **run live in production** a KOSPI 200 index-arbitrage (basket-vs-futures) execution engine: arming and position-sizing logic, fail-closed pre-trade risk gates, automated quote re-pricing, and full order-blotter reconciliation.
- Optimized the firm's **FEP exchange-gateway relay** with the Infrastructure division — TCP-stack tuning and **kernel-bypass** on spread orders — for faster, steadier order delivery (quoting latency ~50 μs typical, best-case ~25 μs).
- On the ongoing **DMA provisional-ledger (가원장) FEP** initiative, designed proposed **API endpoints and the ledger ↔ FEP synchronization algorithm** — letting the OMS fire orders without delay while the gateway validates available balance on every stock order.
- Partner with the firm's **Information Security, IT Strategy, and Infrastructure divisions** (정보보호부 · IT전략부 · 인프라부) on resilient infrastructure and a regular hardware-refresh cycle for uninterrupted uptime; deliver features end to end across a development ↔ production workflow under an independent code-review gate, reporting progress to desk management.

Supergate Inc. · Seoul Summers 2022 & 2023

Software Developer / Research Intern — Applied Cryptography

- Researched and engineered a searchable database over **fully homomorphic encryption (FHE)** — arbitrary computation on encrypted data — in C++, supporting private exact-match and range queries without decryption.
- Co-invented two private-query protocols and filed **two Korean patent applications** ("Encrypted Reverse Index" and "PDQ Protocol over Finite Extension Fields"); the protocols cut query-result size **~50% on average**.
- Cross-compiled the database server and dependencies onto a **Xilinx Zynq FPGA** board for hardware-accelerated encrypted search.
- Directly relevant to digital assets: privacy-preserving computation, encrypted indices, and confidential query protocols underpin tokenized-asset custody, confidential settlement, and on-chain privacy.*

Daangn Market (Karrot) · Seoul 2021

Global Business Development & Product Intern

- Drove international growth of the "Karrot" marketplace in Toronto, helping grow monthly active users **~1.5x**.
- Built a live in-app behavioral analytics dashboard (BigQuery + Looker/Data Studio) used to steer product decisions.
- Ran product-market-fit research and competitive analysis (design, retention, go-to-market) to shape a market-penetration strategy.

University of Pennsylvania · Philadelphia, PA 2021 – 2023

Head Teaching Assistant — Operating Systems (CIS 3800 / MCIT 5950)

- Led and coordinated the teaching-assistant team for 250+ students; ran recitations, code reviews, and office hours.
- Automated edge-case discovery in the homework auto-grader (Python), improving grading reliability.

Unbounded Research in Biomedical Systems Lab · Philadelphia, PA 2019

Machine-Learning Research Assistant

- Built an ensemble of Relief-based feature-selection algorithms (scikit-learn) achieving a **~1.5x speedup** on large biomedical datasets.

SELECTED PROJECTS

Self-Hosted Multi-Asset Portfolio & Finance Platform — Python/FastAPI · HTMX · TimescaleDB · DuckDB

- Real-time personal portfolio and finance platform spanning Korean equities, US equities/FX, and **crypto / digital assets**, with live feeds (KIS OpenAPI, Binance futures, yfinance), server-sent-event push, PnL, and net-worth history.

- Engineered a stateless, **computed-from-source** debt model so monthly card debt clears itself with zero manual entry and cannot drift — a single-source-of-truth design.

AI-Augmented Engineering Harness — applied LLM / multi-agent systems

- How I now build large systems: a multi-agent harness I **direct to design, implement, and review** code — parallel-agent review with adversarial refutation and an independent second-model cross-check, end-to-end feature pipelines, and a self-maintaining decision / knowledge base. Engineering judgment, applied at AI throughput.
- Runs on a self-hosted compute mesh (Tailscale) — Mac orchestrator, always-on mini, wake-on-LAN Linux server — with git-backed sync and scheduled jobs.

Fully Secure Online Voting Platform using FHE — C++ · HElib · Senior Design, "Most Technically Challenging" award

- Led an end-to-end encrypted election system in which ballots are tallied entirely under encryption.

EDUCATION

University of Pennsylvania — School of Engineering & Applied Science · Philadelphia, PA

- **M.S.E., Computer & Information Science** — May 2024. Thesis: *Efficient Private Query Protocol using Fully Homomorphic Encryption*.
- **B.S.E., Computer & Information Science**, Minor in Mathematics — 2023. GPA 3.64/4.00; Dean's List (2021–2023); CIS Senior Design "Most Technically Challenging" award.

PATENTS (FILED)

- "Encrypted Reverse Index" — Korean patent application **10-2023-0115045**.
- "PDQ Protocol Using Finite Extension Fields" — Korean patent application **10-2023-0115051**.

EXPERTISE

Capital Markets & Trading Systems: market making / liquidity provision, index arbitrage, derivatives hedging, market data, exchange (KRX / FEP) connectivity, kernel-bypass networking, low-latency execution, OMS & ledger synchronization, pre-trade risk.

Digital Assets & Cryptography: fully homomorphic encryption, privacy-preserving protocols, crypto / digital-asset data & PnL systems, applied security.

Engineering: C++20, C# / .NET 8, Python, TypeScript / React, SQL · Linux & Windows, DevExpress / WinForms · concurrency & lock-free systems, distributed & real-time architecture · FastAPI / HTMX, PostgreSQL / TimescaleDB · applied LLM / agentic systems.

Architecture, Product & Delivery: system architecture, AI-augmented delivery, product management, data-driven growth, technical-to-business translation, cross-functional leadership.

Languages: Korean (native), English (fluent — U.S. educated).